CHITKARA UNIVERSITY

Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology		
Department Name	Department of Computer Science and Engineering		
Programme Name	B.E (Computer Science & Engineering)		
Course Name	Algorithm Design & Implementation	Session	2026-2027
Course Code	24CSE0317	Semester/Batch	5th/2024
L-T-P (Per Week)	2-0-4	Course Credits	04
Pre-requisite	Fundamentals of C Programming (23CS003), Data Structures (23CS004)	NHEQF Level[1]	5.5
Course Coordinator	Dr. Astha Gupta	SDG Number[4]	4,8,9

1. Objectives of the Course

The scope of the course is to provides the foundation for understanding the key aspects of java programming and implementation obtaining a theoretical understanding of advance programming concepts. The objectives of the course are:

- to build an understanding of analysing and evaluating algorithm efficiency.
- to inculcate the skill in students to implement and optimize data structures.
- to develop and apply advanced sorting and searching algorithms.
- to solve complex problems using greedy algorithms, backtracking, and dynamic programming.

2. Course Learning Outcomes

After completion of the course, the student should be able to:

	Course Learning Outcome	*POs	NHEQF Level Descriptor[1]	Sessions
CLO01	Understand and apply the concept of algorithm complexity	PO1, PO2, PO3, PO4, PO5, PO11	Q1, Q2	16
CLO02	Proficiency in implementing hash tables, heaps, priority queues	PO1, PO2, PO3, PO4, PO5, PO8, PO11	Q2, Q3	22
CLO03	Implement the advanced sorting algorithms and apply appropriate algorithm for a particular problem	PO1, PO2, PO3, PO4, PO5, PO8, PO11	Q2, Q3, Q4	24
CLO04	Apply the concepts and theories of different algorithmic strategies	PO1, PO2, PO3, PO4, PO5, PO8, PO11	Q1, Q2, Q4	28
Total Contact Hours				**90**

CLO-PO Mapping

Course Learning Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO01	PSO02	Type of Assessment's3
CLO01	H	H	L	M	M	-	-	-	-	-	M	H	M	Formative, Summative
CLO02	H	M	M	L	H	-	-	L	-	-	M	H	H	Formative, Summative
CLO03	H	H	H	M	H	-	-	L	-	-	M	H	H	Formative, Summative
CLO04	H	H	H	M	M	-	-	L	-	-	H	H	M	Formative, Summative

H=High, M=Medium, L=Low

Revised Bloom's Taxonomy Terminology
* PO's available at (shorturl.at/cryzF)
**Cognitive Level =CL
***Knowledge Categories = KC

[2] Refer to Annexure

3. **Recommended Books:**

Textbooks:

B01: "Classic Data Structures", Samanta and Debasis, 2nd edition, PHI publishers.
B02: "Fundamentals of Data Structures", Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
B03: "Design and Analysis of Algorithms" by S. Sridhar, Oxford University Press, 2014.

Reference Books:

B04: "Data Structures with C (Schaum's Outline Series)", Seymour Lipschutz, 1st edition, McGraw Hill Education.
B05: "Algorithms, Data Structures, and Problem Solving with C++", Illustrated Edition by Mark Allen Weiss, Addison-Wesley Publishing Company.

E-Resources:

- https://library.chitkara.edu.in/subscribed-books.php

4. **Other readings and relevant websites:**

S. No.	Link of Journals, Magazines, websites and Research Papers
1.	https://www.freecodecamp.org/news/big-o-cheat-sheet-time-complexity-chart
2.	https://www.geeksforgeeks.org/distributing-m-items-circle-size-n-starting-k-th-position/
3.	https://www.geeksforgeeks.org/open-addressing-collision-handling-technique-in-hashing/
4.	https://www.geeksforgeeks.org/find-the-winner-of-the-game-2/
5.	https://www.geeksforgeeks.org/priority-queue-using-linked-list/
6.	https://www.youtube.com/watch?v=HqPJF2L5h9U

[1] National Higher Education Qualification Framework Level, Refer to annexure

[1] NHEQF Level Descriptor, Refer to Annexure & Learning outcomes descriptors for qualification for all levels on the NHEQF
[3]Types of Assessments can be referred from Type of Assessments. Refer to Annexure.
[4]For SDG Mapping with Courses, Pl refer SDG Mapping policy for Courses

5. **Recommended Tools and Platforms**
Test Pad

6. **Course Plan (Theory + Lab Plan):**

Theory Plan

Lecture Number	Topics	TextBooks
1-2	**Unit-1: Overview of ADS, Time and Space Complexity** – Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II. Practice Problem: Prime Factorization, GCD of two numbers, Distribute in circle.	BO1-Chapter 1, BO4- Chapter 1, BO5- Chapter 2
3-5	**Unit-2: Arrays & Strings** – Array Prefix/Suffix, Difference Array, Sliding Window, Two Pointers, Binary Search on Array, Array Rotation, In-place Reversal, Partitioning, String Basics: Reverse, Palindrome, Anagram, Substrings, String Pattern Matching, KMP Basics, Z-Function Intro, String Manipulation: Compression, Hashing, Rolling Hash, Greedy on Arrays: Activity Selection, Interval Problems, DP on Arrays: LIS, LCS, 1D DP patterns, Practice Contest 2: 3 medium array/string problems, Practice Contest 3: 3 mixed array/string problems.	BO2- Chapter 2, BO3- Chapter 2, BO5- Chapter 3
6-10	**Unit-3: Greedy Algorithms** – Greedy Thinking, Proof Techniques, Exchange Arguments, Interval Scheduling, Meeting Rooms, Task Scheduling, Greedy on Arrays: Coins, Fractions, Min/Max Problems, Greedy on Strings: Lexicographical Min/Max, Reordering, Greedy with Sorting, Priority Queues, Greedy + DP Hybrid Problems, Practice Contest 4: 3 greedy-focused problems, Practice Contest 5: 3 mixed greedy + array/string problems.	BO4-Chapter 8, BO5- Chapter 10
11-15	**Unit-4: Stack & Queue** – Stack Basics, Monotonic Stack, Next Greater Element, Stack Applications: Parentheses, Expression Evaluation, Queue Basics, Circular Queue, Deque, Sliding Window Maximum with Deque, Stack + Greedy, Stack + DP Patterns, Practice Contest 6: 3 stack/queue problems.	BO1-Chapter 3 & 4, BO2- Chapter 3 & 4, BO3-Chapter 5 & 6

[2] Refer to Annexure

16-19	**Unit-5: Hashing & Mapping** – Hash Maps, Hash Sets, Collision Handling, Frequency Counting, Two Sum Variants, Subarray Sum Equals K, Subarray with Constraints, String Hashing, Rolling Hash, Substring Search, Coordinate Compression, Map-Based DP, Practice Contest 7: 3 hashing/mapping problems.	BO1-Chapter 9, BO5-Chapter 5
ST-1 (Syllabus covered from Lecture 1 to 19)		
20-22	**Unit-6: Recursion & Backtracking** – Recursion Basics, Recursion Tree, Depth-First Search, Backtracking Patterns: Subsets, Permutations, Combinations, N-Queens, Sudoku Solver, Maze Problems, Backtracking with Pruning, Optimization, Backtracking + DP (Memoization) Intro, Practice Contest 8: 3 backtracking problems, Practice Contest 9: 3 mixed recursion/backtracking problems.	BO4-Chapter 4 & 5, BO5-Chapter 10
23-25	**Unit-7: Trees** – Tree Basics, Traversals: DFS (Pre, In, Post), BFS, Recursive Tree Problems: Height, Diameter, LCA, Binary Search Tree: Search, Insert, Delete, Validation, BST Queries: K-th Smallest, Range Sum, Floor/Ceil, Tree DP: Max Independent Set, Path Sum, Diameter DP, Trie Basics: Insert, Search, Prefix Problems, Trie + DP, Trie + Greedy, Practice Contest 10: 3 tree problems, Practice Contest 11: 3 mixed tree + DP.	BO1-Chapter 6 & 7, BO2-Chapter 5, BO5-Chapter 4
26-28	**Unit-8: Heap** – Introduction to Heap, Min Heap, Max Heap, Operations on Heap, Priority Queues, Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, check if a given tree is max-heap or not.	BO1-Chapter 8, BO2-Chapter 8, BO5-Chapter 6
29-30	**Unit-9: Graphs** – Graph Basics: Adjacency List, BFS, DFS, Connected Components, Shortest Path: BFS for unweighted, Dijkstra, Minimum Spanning Tree: Kruskal, Prim, Union-Find, Topological Sort, DAG Problems, DP on DAG, Graph DP: Paths, Counting, Shortest Path DP, Bipartite Checking, Coloring, Cycle Detection.	BO1-Chapter 10, BO4-Chapter 9, BO5-Chapter 9
ST-2 (Syllabus covered from Lecture 20 to 30)		
END TERM – FULL SYLLABUS		

Lab Plan

Lab Number	Practical
1-6	**Unit-1: Introduction to Time and Space Complexity:** 1. Find the prime factors of a given number. 2. Compute GCD of two numbers using Euclidean Algorithm. 3. Distribute N items in a circle and find who gets the last one. 4. Solve a problem using brute force. 5. Give one example of a randomized algorithm. 6. Use divide and conquer to sort an array.
7-17	**Unit-2: Arrays & Strings** 1. Find a pair with sum K using two pointers. 2. Find the maximum sum subarray of size *k* using the sliding window technique. 3. Rotate an array using the reversal algorithm. 4. Check whether two strings are anagrams. 5. Find the Longest Common Subsequence (LCS). 6. Find the Longest Increasing Subsequence (LIS) using Dynamic Programming.
18-25	**Unit-3: Greedy Algorithms** 1. Apply the greedy approach to interval scheduling. 2. Solve the fractional knapsack problem. 3. Schedule jobs with deadlines to maximize profit. 4. Solve the activity selection problem using the greedy method.
26-32	**Unit-4: Stack & Queue** 1. Implement stack operations using an array. 2. Check balanced parentheses using a stack. 3. Implement circular queue operations.
33-38	**Unit-5: Hashing & Mapping** 1. Solve the "Noise in the Library" problem using hashing. 2. Use hashing to balance a scale with given weights. 3. Find the winner in a voting system using hashing.

[2] Refer to Annexure

39-45	**Unit-6: Recursion & Backtracking** 1. Solve the Tower of Hanoi using recursion. 2. Generate permutations of a string using backtracking. 3. Move a robot into a grid using backtracking. 4. Solve Sudoku using backtracking. 5. Find all paths for Rat in a Maze. 6. Print all binary strings of *n* bits. 7. Solve the N-Queens problem using backtracking. **Dynamic Programming Applications:** 8. Solve the 0/1 Knapsack problem using Dynamic Programming. 9. Count the number of ways to cover a distance of *N* using 1, 2, or 3 steps. 10. Solve Matrix Chain Multiplication using Dynamic Programming.
46-51	**Unit-7: Trees** 1. Perform preorder, inorder, and postorder traversals of a binary tree. 2. Find the height of a binary tree. 3. Find the Lowest Common Ancestor (LCA) of two nodes in a Binary Search Tree. 4. Find the kth smallest element in a Binary Search Tree.
52-56	**Unit-8: Heap** 1. Find the maximum/minimum element in a continuous stream of data. 2. Sort an array using Heap Sort. 3. Check whether a given tree is a Max-Heap.
57-60	**Unit-9: Graphs** 1. Find the cycle in an undirected graph. 2. Find the minimum number of edges in a path of a graph. 3. Find a path in a directed graph. 4. Find the number of islands. 5. Find the shortest path using Dijkstra's algorithm. 6. Solve a graph with negative edge weights using the Bellman-Ford algorithm.

7. **Delivery/Instructional Resources**

Lecture No.	Topics	CLO	Book No, CH No, Page No	TLM [1]	ALM [2]	Web References	Audio-Video
1-2	**Unit-1: Overview of ADS, Time and Space Complexity** – Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II. Practice Problem: Prime Factorization, GCD of two numbers, Distribute in circle.	CLO01	BO1-Chapter 1, BO4-Chapter 1, BO5-Chapter 2	Lecture Discussion	Quiz, Challenge, Assignment	https://www.freecodecamp.org/news/big-o-cheat-sheet-time-complexity-chart	https://www.youtube.com/watch?v=HfIH3czXc-8
3-5	**Unit-2: Arrays & Strings** – Array Prefix/Suffix, Difference Array, Sliding Window, Two Pointers, Binary Search on Array, Array Rotation, In-place Reversal, Partitioning, String Basics: Reverse, Palindrome, Anagram, Substrings, String Pattern Matching, KMP Basics, Z-Function Intro, String Manipulation: Compression, Hashing, Rolling Hash, Greedy on Arrays: Activity Selection, Interval Problems, DP on Arrays: LIS, LCS, 1D DP patterns, Practice Contest 2: 3 medium array/string problems, Practice Contest 3: 3 mixed array/string problems.	CLO03, CLO04	BO2-Chapter 2, BO3-Chapter 2, BO5-Chapter 3	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/	https://www.youtube.com/watch?v=RBSGKlAvoiM

[2] Refer to Annexure

6-10	**Unit-3: Greedy Algorithms** – Greedy Thinking, Proof Techniques, Exchange Arguments, Interval Scheduling, Meeting Rooms, Task Scheduling, Greedy on Arrays: Coins, Fractions, Min/Max Problems, Greedy on Strings: Lexicographical Min/Max, Reordering, Greedy with Sorting, Priority Queues, Greedy + DP Hybrid Problems, Practice Contest 4: 3 greedy-focused problems, Practice Contest 5: 3 mixed greedy + array/string problems.	CLO02, CLO04	BO4-Chapter 8, BO5-Chapter 10	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/greedy-algorithms/	https://www.youtube.com/watch?v=ARvQcqJ_-NY
11-15	**Unit-4: Stack & Queue** – Stack Basics, Monotonic Stack, Next Greater Element, Stack Applications: Parentheses, Expression Evaluation, Queue Basics, Circular Queue, Deque, Sliding Window Maximum with Deque, Stack + Greedy, Stack + DP Patterns, Practice Contest 6: 3 stack/queue problems.	CLO03	BO1-Chapter 3 & 4, BO2-Chapter 3 & 4, BO3-Chapter 5 & 6	Lecture Discussion	Quiz, Challenge, Assignment		https://www.youtube.com/watch?v=wjI1WNcIntg
16-19	**Unit-5: Hashing & Mapping** – Hash Maps, Hash Sets, Collision Handling, Frequency Counting, Two Sum Variants, Subarray Sum Equals K, Subarray with Constraints, String Hashing, Rolling Hash, Substring Search, Coordinate Compression, Map-Based DP, Practice Contest 7: 3 hashing/mapping problems.	CLO02, CLO04	BO1-Chapter 9, BO5-Chapter 5	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/string/string-hashing.html	https://www.youtube.com/watch?v=shs0KM3wKv8
20-22	**Unit-6: Recursion & Backtracking** – Recursion Basics, Recursion Tree, Depth-First Search, Backtracking Patterns: Subsets, Permutations, Combinations, N-Queens, Sudoku Solver, Maze Problems, Backtracking with Pruning, Optimization, Backtracking + DP (Memoization) Intro, Practice Contest 8: 3 backtracking problems, Practice Contest 9: 3 mixed recursion/backtracking problems.	CLO04	BO4-Chapter 4 & 5, BO5-Chapter 10	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/algebra/all-submasks.html	https://www.youtube.com/watch?v=DKCbsiDBN6c
23-25	**Unit-7: Trees** – Tree Basics, Traversals: DFS (Pre, In, Post), BFS, Recursive Tree Problems: Height, Diameter, LCA, Binary Search Tree: Search, Insert, Delete, Validation, BST Queries: K-th Smallest, Range Sum, Floor/Ceil, Tree DP: Max Independent Set, Path Sum, Diameter DP, Trie Basics: Insert, Search, Prefix Problems, Trie + DP, Trie + Greedy, Practice Contest 10: 3 tree problems, Practice Contest 11: 3 mixed tree + DP.	CLO02, CLO04	BO1-Chapter 6 & 7, BO2-Chapter 5, BO5-Chapter 4	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/graph/lca.html	https://www.youtube.com/watch?v=fAAZixBzIAI

[2] Refer to Annexure

26-28	**Unit-8: Heap** – Introduction to Heap, Min Heap, Max Heap, Operations on Heap, Priority Queues, Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, check if a given tree is max-heap or not.	CLO02, CLO03	BO1-Chapter 8, BO2-Chapter 8, BO5-Chapter 6	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/algebra/all-submasks.html	https://www.youtube.com/watch?v=DKCbsiDBN6c
29-30	**Unit-9: Graphs** – Graph Basics: Adjacency List, BFS, DFS, Connected Components, Shortest Path: BFS for unweighted, Dijkstra, Minimum Spanning Tree: Kruskal, Prim, Union-Find, Topological Sort, DAG Problems, DP on DAG, Graph DP: Paths, Counting, Shortest Path DP, Bipartite Checking, Coloring, Cycle Detection.	CLO04	BO1-Chapter 10, BO4-Chapter 9, BO5-Chapter 9	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/backtracking-algorithms/	https://www.youtube.com/watch?v=QXkY2DOUCWM

8. Remedial Classes[4]

Slow Learners	Average Learners	Fast Learners
• Remedial Classes on Saturdays • Encouragement for improvement using Peer Tutoring • Use of Audio and Visual Materials • Use of Real-Life Examples	• Workshops • Formative Exercises used to highlight concepts and notions • E-notes and E-exercises to read ahead of the pedagogic material.	• Engaging students to hold hands of slow learners by creating a Peer Tutoring Group • Design solutions for complex problems • Design Solutions for complex problems • Presentation on topics beyond those covered in CHO

9. Self-Learning[2]

Assignments to promote self-learning, a survey of contents from multiple sources.

S. No.	Topics	CLO	ALM	References/MOOCS
1	Algorithm Analysis & Complexity	CLO01	Group Discussion / Coding Practice	https://nptel.ac.in/courses/106105164
2	Arrays, Strings & Searching Techniques	CLO03	Group Discussion / Coding Practice	https://www.coursera.org/learn/cpsc-8400-design-and-analysis-of-algorithms
3	Stack, Queue & Priority Queue Applications	CLO02	Group Discussion / Coding Practice	https://nptel.ac.in/courses/106106145
4	Hashing & String-Matching Algorithms	CLO02	Group Discussion / Coding Practice	https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/
5	Heaps & Priority Queues	CLO02	Group Discussion / Coding Practice	https://www.coursera.org/learn/algorithms-part1
6	Trees, BST & Trie	CLO02	Group Discussion / Coding Practice	https://www.coursera.org/learn/algorithms-part1

10. Delivery Details of Content Beyond Syllabus[3]

S. No.	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS

[2] Refer to Annexure

1.	Advanced Data Structures: Segment Tree & Fenwick Tree	CLO02	PO1, PO2	Programming Assignment	https://cp-algorithms.com/data_structures/segment_tree.html
2.	String Algorithms: Rabin–Karp, Aho–Corasick	CLO02	PO1, PO2, PO3	Problem-Based Learning	https://cp-algorithms.com/string/
3.	Dynamic Programming Optimization (Bitmask DP, State DP)	CLO04	PO1, PO2, PO3	Problem-Based Learning	https://cp-algorithms.com/dynamic_programming/

11. Evaluation Scheme & Components:

Assessment Type[4]	Evaluation Component[5]	Type of Component[6]	No. of Assessments[7]	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative, Summative	Test Pad Coding Series	Component 1	01	10%	10	Online/ Off Campus	CLO01 to CLO04
Formative, Summative	Formal Assessment (FAs)		05*	10%	10	Online/ On Campus	CLO01 to CLO04
Formative, Summative	Sessional Tests (STs)	Component 2	02*	30%	30	Online/ On Campus	CLO01 to CLO04
Summative	End Term Examination	Component 3	01	50%	50	Online/ On Campus	CLO01 to CLO04

* Out of 05 FAs, best 4 FA for final marks evaluation of FAs will be considered.

* Out of 02 STs, best 1 ST for final marks evaluation of STs will be considered.

* Makeup Examination will compensate for either ST-1 or ST-2 (Only for genuine cases, based on the Dean's approval and the exam will be based on the full syllabus).

**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

[3] Refer to Annexure

[4] Refer to Annexure 2 of NCrF

[5] Refer to Annexure

[6] Refer to Annexure

[2] Refer to Annexure

12. Syllabus of the Course:

Subject: Algorithm Design & Implementation	**Subject Code:** 24CSE0317	
Topic(s)	**No. of Sessions**	**Weightage (%)**
Unit-1: Overview of ADS, Time and Space Complexity – Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II. Practice Problem: Prime Factorization, GCD of two numbers, Distribute in circle.	6	7%
Unit-2: Arrays & Strings – Array Prefix/Suffix, Difference Array, Sliding Window, Two Pointers, Binary Search on Array, Array Rotation, In-place Reversal, Partitioning, String Basics: Reverse, Palindrome, Anagram, Substrings, String Pattern Matching, KMP Basics, Z-Function Intro. String Manipulation: Compression, Hashing, Rolling Hash, Greedy on Arrays: Activity Selection, Interval Problems, DP on Arrays: LIS, LCS, 1D DP patterns, Practice Contest 2: 3 medium array/string problems, Practice Contest 3: 3 mixed array/string problems	15	17%
Unit-3: Greedy Algorithms – Greedy Thinking, Proof Techniques, Exchange Arguments, Interval Scheduling, Meeting Rooms, Task Scheduling, Greedy on Arrays: Coins, Fractions, Min/Max Problems. Greedy on Strings: Lexicographical Min/Max, Reordering, Greedy with Sorting, Priority Queues, Greedy + DP Hybrid Problems, Practice Contest 4: 3 greedy-focused problems, Practice Contest 5: 3 mixed greedy + array/string problems.	12	13%
Unit-4: Stack & Queue – Stack Basics, Monotonic Stack, Next Greater Element, Stack Applications: Parentheses, Expression Evaluation, Queue Basics, Circular Queue, Deque, Sliding Window Maximum with Deque, Stack + Greedy, Stack + DP Patterns, Practice Contest 6: 3 stack/queue problems.	10	11%
Unit-5: Hashing & Mapping – Hash Maps, Hash Sets, Collision Handling, Frequency Counting, Two Sum Variants, Subarray Sum Equals K, Subarray with Constraints. String Hashing, Rolling Hash, Substring Search, Coordinate Compression, Map-Based DP, Practice Contest 7: 3 hashing/mapping problems.	11	12%
ST-1 (Covering 60% Syllabus)		
Unit-6: Recursion & Backtracking – Recursion Basics, Recursion Tree, Depth-First Search, Backtracking Patterns: Subsets, Permutations, Combinations, N-Queens, Sudoku Solver, Maze Problems.	11	12%
Unit-7: Trees – Tree Basics, Traversals: DFS (Pre, In, Post), BFS, Recursive Tree Problems: Height, Diameter, LCA, Binary Search Tree: Search, Insert, Delete, Validation, BST Queries: K-th Smallest, Range Sum, Floor/Ceil. Tree DP: Max Independent Set, Path Sum, Diameter DP, Trie Basics: Insert, Search, Prefix Problems, Trie + DP, Trie + Greedy, Practice Contest 10: 3 tree problems, Practice Contest 11: 3 mixed tree + DP.	12	13%
Unit-8: Heap – Introduction to Heap, Min Heap, Max Heap, Operations on Heap, Priority Queues, Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, check if a given tree is max-heap or not.	6	7%
Unit-9: Graphs – Graph Basics: Adjacency List, BFS, DFS, Connected Components, Shortest Path: BFS for unweighted, Dijkstra, Minimum Spanning Tree: Kruskal, Prim, Union-Find, Topological Sort, DAG Problems, DP on DAG, Graph DP: Paths, Counting, Shortest Path DP, Bipartite Checking, Coloring, Cycle Detection.	7	8%
ST-2 (Covering 40% Syllabus)		
End Term 100% Syllabus		

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Astha Gupta	
Head-Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date	01/07/2026	